DATA ANALYST PROJECT · SQL + PYTHON

AI-Powered Demand Forecasting for Retail

Turning two years of sales data into inventory and marketing decisions

THE CHALLENGE

Retail Loses Money When Demand Is a Guess

1. **Overstock**

   Cash tied up in unsold inventory, forced markdowns, and waste.

2. **Understock**

   Lost sales exactly when demand peaks — the worst time to run out.

3. **Untimed Marketing**

   Broad campaigns waste spend when they aren't timed to real demand shifts.

Can we forecast near-term sales accurately enough — by store and category — to guide inventory and marketing decisions?

APPROACH

How We Solved It

1 Structure the Data

Loaded 2 years of daily sales into SQL

2 Explore Patterns

Queried trends, seasonality, and promo effects

3 Engineer Features

Built lag & rolling features via SQL window functions

4 Model & Compare

Baselines, SARIMA, and XGBoost

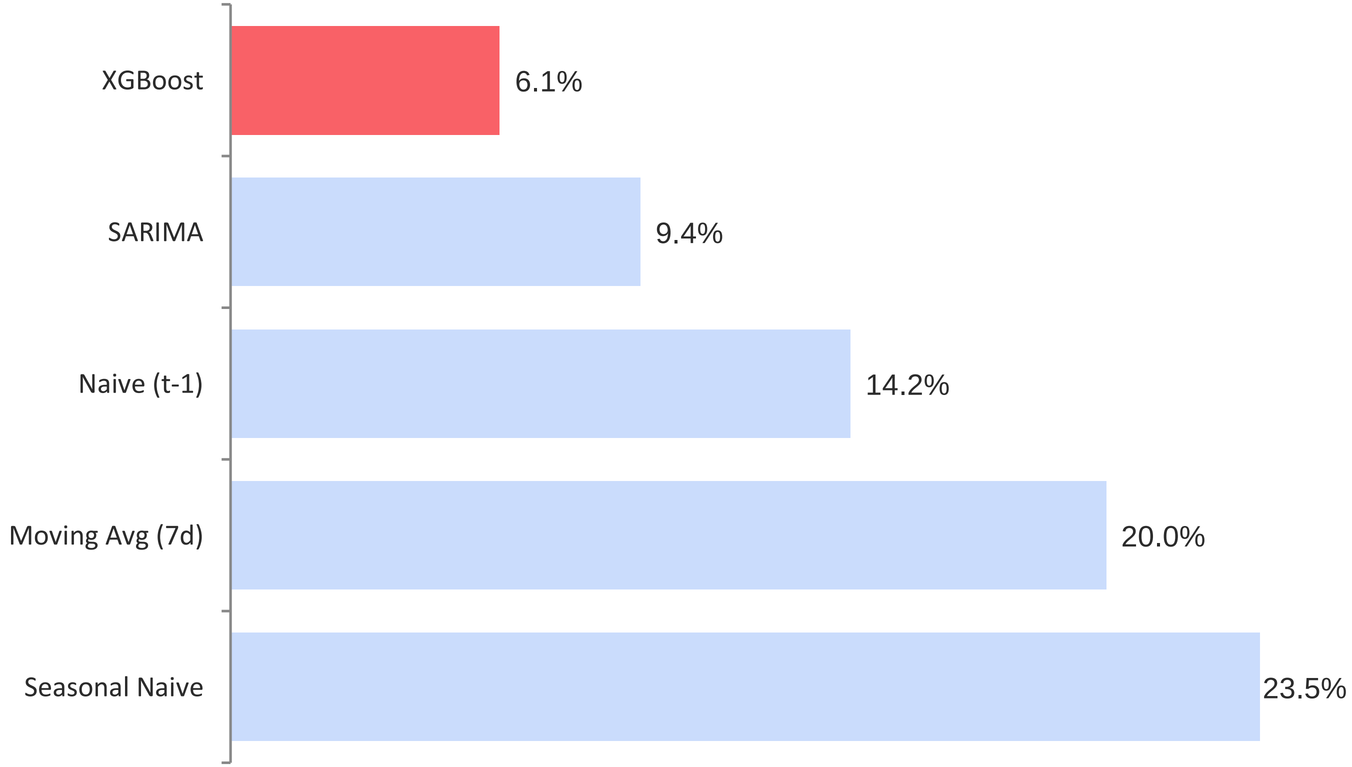
5 Evaluate Rigorously

Tested on the hardest period: Black Friday-Christmas

RESULTS

More Than 2x More Accurate

Forecast Error by Model (MAPE, lower = better)



6.1%

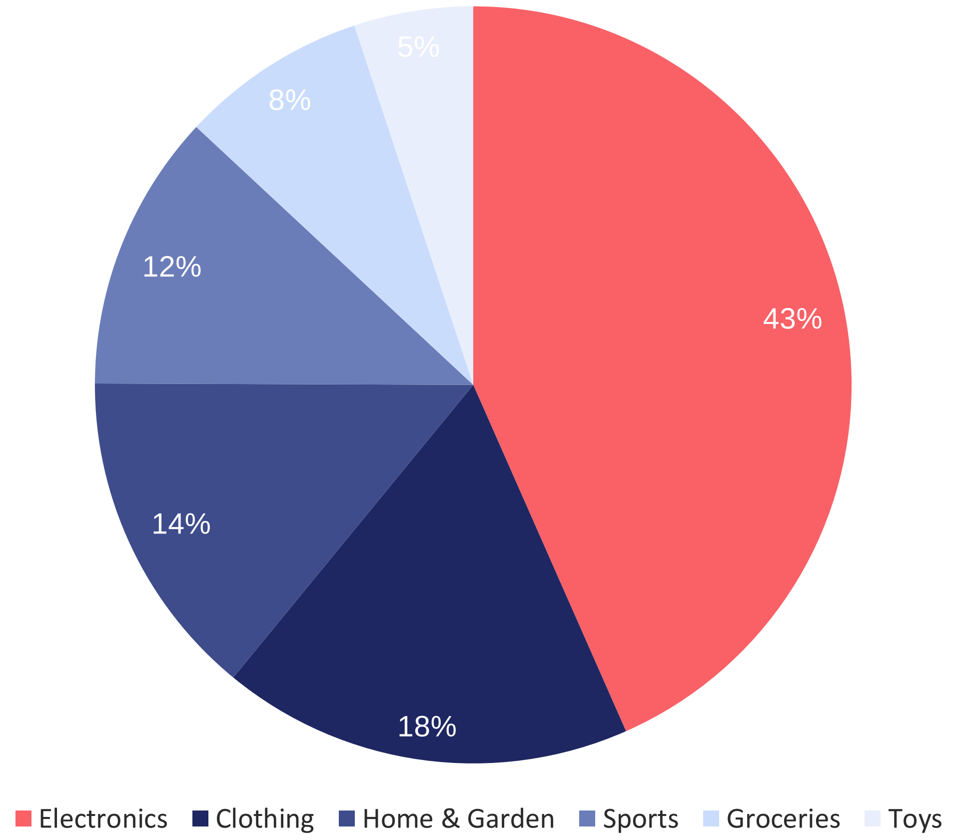
average forecast error (XGBoost) on the hardest test window — Nov-Dec holidays

2.3x

more accurate than a naive “yesterday repeats” forecast

KEY INSIGHT

Electronics Drives Almost Half of Revenue



43%

of total revenue comes from Electronics

Despite moderate unit volume, Electronics is the highest-leverage category for forecast accuracy — a 1% error here costs more in dollar terms than anywhere else.

KEY INSIGHT

Four Signals That Move the Needle

2.3x

Holiday revenue lift

vs. an average day — the single strongest signal in the forecast model (72% of feature importance)

+43%

Promotion lift

increase in units sold on promo days vs. non-promo days

+18%

Weekend lift

higher average daily revenue on Sat/Sun vs. weekdays

22%

Store performance range

top vs. bottom store — consistent enough for one shared forecasting model

WHAT THIS MEANS FOR THE BUSINESS

Five Recommendations

1. **Prioritize Electronics forecast accuracy**

   It's 43% of revenue — the highest dollar impact of any category.

2. **Build holiday calendars into inventory planning**

   Not just historical averages — holidays are the strongest demand signal found.

3. **Plan stock around promotions**

   Treat promos as a measured ~43% demand lever, not a guess.

4. **Set a standing weekend adjustment**

   +18% is consistent enough to bake into staffing and stocking by default.

5. **Use one shared model across stores**

   Performance is even enough (22% range) that five separate models aren't needed.

WHERE THIS GOES NEXT

- Forecast per store × category, not just the aggregate total — the data already supports it
- Add Facebook Prophet, using holidays as explicit regressors
- Tune SARIMA orders with a grid search instead of a fixed configuration

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From guesswork to an evidence-based forecast.